

Marco Chen

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EDUCATION

The Cooper Union for the Advancement of Science and Art

New York, NY

B.E. in Electrical Engineering; GPA: 3.43

Expected May 2028

Relevant Coursework: Operating Systems, Computer Architecture, Digital Signal Processing, Digital Logic Design

EXPERIENCE

Linux Kernel Mentorship Program, The Linux Foundation

June 2026 – Dec 2026

Industrial I/O (IIO) Subsystem Contributor

Remote

- Found and fixed a NULL-pointer dereference in the mainline MAX30102 IIO driver, where the threaded IRQ handler dereferenced `active_scan_mask` outside buffered mode; accepted after four rounds of maintainer review
- Reproduced the crash by bringing up a MAX30102 over I2C on a cross-compiled arm64 Raspberry Pi 4 kernel, writing the device tree overlay including the pull-up required by its open-drain interrupt line
- Implementing runtime power management for the Bosch BMA400 accelerometer driver with the kernel runtime PM framework, suspending the sensor between accesses; drafting the design proposal for the linux-iio mailing list

Cooper Union Autonomy Lab (IGVC)

Sept 2025 – Present

Firmware Team Lead

New York, NY

- Built a FreeRTOS+CLI debug shell over UART0 on RP2350, exposing commands to list tasks and change task priorities at runtime without halting the target under a debugger
- Documented an SWD debugging toolchain for RP2350 using an RP2040 as a CMSIS-DAP probe with OpenOCD and ARM-GDB, with setup guides for Linux, macOS, and Windows

Cooper Union Summer STEM Program

June 2025 – Aug 2025

Embedded Systems Teaching Assistant

New York, NY

- Taught embedded C to high school students through worked examples on circular buffers, bit manipulation, I2C, and interrupts, and advised student teams on device drivers for their final autonomous-system projects
- Set up and documented an AVR-GDB debugging toolchain over debugWIRE across Linux, macOS, and Windows, moving students from the Arduino IDE to command-line build and debug workflows

PROJECTS

OTA Firmware Update System (A/B Bootloader) | C, STM32L4, Linker Scripts

Aug 2026 – Present

- Building the device half of a two-person firmware update system: a stage-2 bootloader and linker scripts placing two application slots in STM32L4 dual-bank flash, with manual vector table and stack handoff
- Co-designing the update protocol and slot metadata: chunked UART transfer with per-chunk CRC32, resume by chunk index, and a validity marker written only after full-image verification so power loss is survivable

MCP2518FD SPI-to-CAN Driver (CAN 2.0) | Embedded C, RP2350, CAN bus, SPI

Feb 2026 – Apr 2026

- Built a CAN 2.0 driver in C for the MCP2518FD on RP2350 from the MCP25XXFD datasheets, with a HAL for SPI register access and an API for sending, receiving, and configuring CAN messages
- Debugged an initialization hang with a logic analyzer, tracing it to SPI reads and writes that never addressed the peripheral, then verified CAN traffic on an oscilloscope and against a USB-to-CAN adapter with can-utils

STM32 Health Wearable | Independent Study, C, FreeRTOS, STM32L4

June 2026 – July 2026

- Wrote interrupt-driven I2C drivers for a BMI270 IMU and MAX30102 PPG sensor, working around the STM32 HAL's shared I2C callbacks with per-peripheral context structs and semaphores so both sample concurrently
- Implemented a five-state fall-detection state machine that confirms a fall 2.0 s after impact, deriving thresholds from drop tests and writing the results up as an IEEE-format report

Multi-process Synchronization Library | C, GDB, UNIX Signals

Nov 2025 – Dec 2025

- Built spinlock, semaphore, and producer-consumer FIFO libraries over `MAP_SHARED` mmap'd memory using `sigsuspend` sleep/wakeup, debugging a lost-wakeup deadlock from signals arriving before `sigsuspend`

SKILLS

Languages: C, Python, C++, x86 Assembly, Bash, MATLAB

Technologies: FreeRTOS, Linux Kernel (IIO), I2C, SPI, UART, CAN, Linker Scripts, STM32L4, RP2350, ATmega328P

Tools: Git, GDB, OpenOCD, SWD, debugWIRE, QEMU, GNU Make, Saleae Logic Analyzer, Oscilloscope, Linux

Certifications: GIAC Foundational Cybersecurity Technologies (GFACT)